

LLM

Language Model → A statistical model that predicts the next word in a sequence or context

Large Language Models → Advanced LMs trained on massive datasets with billions or trillions of parameters.

Tokens → are often sub words, characters, or punctuation.

Frontier Model: eng o'ziri va eng kuchli modellar

GPT-4 ~ 1.8 trillion parameter

Katta model qurish emas, haqiqatda maksimal foydalanish.

FLOP → hisoblashni o'lchash birligi.

NT hisoblash narxini o'lchashning standart birligi.

FLOP → Bu "AI ning puli". Compute qancha hisoblash kerak.

$$C \sim G \times N \times D$$

Kotta modellarda **Attention Optimization** siyosi deyarli ahamiyatsiz

Emergence → ma'lum bir darajada model da paydo bo'ladigan qobiliyat.

Scaling Laws → model sifati, dataset hajmi, yoki compute bilan quvvat qonuni ko'rinishida qanday yaxshilashni ko'rsatadi

Chirchillo qonuni → optimal o'qitish uchun token soni parametri sonidan taxminan $20 \times$ ko'p bo'lishi kerak

Bitter Lesson → masshtabli umumiy hisoblashda
foydalanadigan AI usullari har doim
qolda yaratilgan bilim yoki "aqilli
fokestarlar" uskun kelgan,

Samaradorlik multiplieri: Texnikalar va
resurslarni yaxshi ishlatish

Belgilangan compute ~~by~~ budget bilan eng
yaxshi modelni qanday qurish mumkin?

Bir vaqtning o'zida bir nechta resursni
balanslash orqali ko'p o'lchovli optimizatsiya
erishish mumkin.

- **Compute (Flop)** → Qancha parametr, koren, batch
- **Data** → Qanday sifat, filtrlar, domenlar
- **Xotira (VRAM)** → Aniglik, Paralelizm.
- **Hardware** → Qaysi TPU/GPU

Attention → feature örgünish

Transformer: faqat attention ga tayanadigan
neuron tarmoq

Self-attention va MLP ketma ket keladi.
Tokenlar **parallel** ishlanadi

~~At~~ Attention → MLP → Tokenizer

Tokenization: raw matnni neuron tarmoq qayta
ishlaydigan 'integer ID ga o'tkazish'

BPE → Byte Pair Encoding sozlarni eng kdp
uchradigan juftlari bilan birlashtiradi.

Alignment → fil modelini inson niyyatlariga
yonaltilish

RLHF → Reinforcement Learning from Human Feedback.